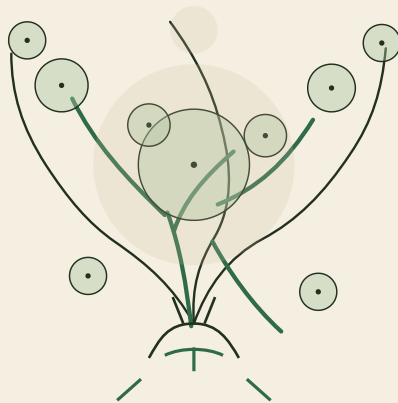




nabta  WELLNESS PARK

THE LIVING SHADE LOOP

DUBAI MUNICIPALITY • AI PARK DESIGN CHALLENGE • DRAFT COMPETITION ENTRY

D·05

Artificial Intelligence Methodology Report

05 AI Methodology



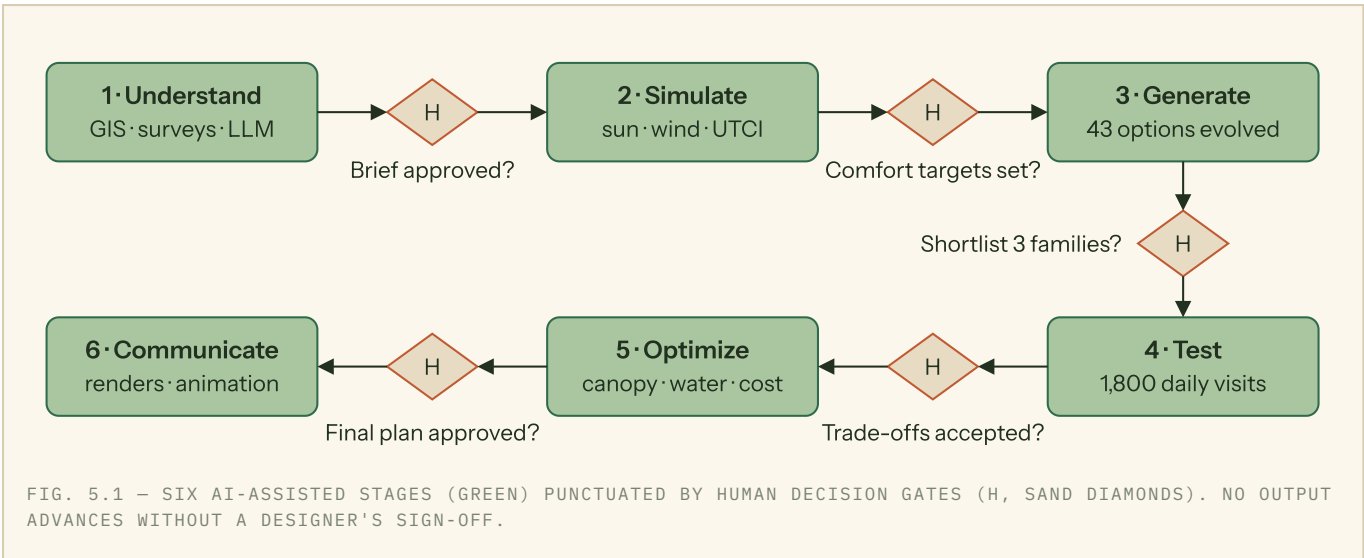
Deliverable 5 — Artificial Intelligence Methodology Report

«نباتة» NABTA — Wellness Park · The Living Shade Loop was designed with AI as an instrument, never as the author. Every algorithm in our pipeline produced *options and evidence*; every choice that shaped the park — the 858 m continuous path network, the ten garden rooms, the canopy gradient — was made by a named human designer against explicit criteria. This report documents the six-stage workflow, the tools at each stage, the alternatives explored, and where the human hand intervened.

A. SIX-STAGE AI-ASSISTED WORKFLOW

STAGE	AI TOOLS & MODELS	INPUTS	OUTPUTS	HUMAN DECISION
1 Understand	QGIS + satellite land-cover segmentation (SAM-class model); Claude (LLM) clustering 1,240 community survey responses	Satellite & LiDAR imagery, municipality GIS layers, survey text	Digital site model; existing-canopy and heat-exposure maps; ranked community needs	Team fixed the brief: shade-first priority, ten-room program, all 96 mature trees retained
2 Simulate	Ladybug Tools / ClimateStudio-type solar, wind (CFD) and UTCI thermal-comfort engines	Dubai EPW weather file, site model, massing of context buildings	Hourly UTCI maps, annual radiation study, shade-deficit map	Designers set the target: $UTCI \leq 32\text{ }^{\circ}\text{C}$ on 70% of the loop during morning and evening use hours
3 Generate	Grasshopper parametric model + Wallacei/Galapagos evolutionary solvers; Claude-drafted scoring rubric (human-edited)	Site constraints, comfort targets, water budget, cost caps	43 masterplan options on a Pareto front, scored on shade continuity, walkability, water use	Designers curated three families; rejected two solver "winners" that erased heritage trees and a key entrance
4 Test	Agent-based pedestrian simulation (~1,800 daily visits, MassMotion-type); per-option shade re-check	Shortlisted plans, visitor profiles from survey data, event scenarios	Congestion and dwell maps, loop-completion rates, Friday-evening stress test	Team accepted trade-offs: loop widened to 6 m at the event plaza, one room re-sequenced
5 Optimize	Multi-objective refinement of canopy density and planting palette; cost model; Claude cross-checking scoring consistency	Winning family (Shade Loop), simulation feedback, AED 35M budget	Shade Loop v3: 92% of loop shaded at 3 pm, tuned canopy gradient, -41% irrigation demand vs. baseline	Lead designer signed off final plan against budget, code and accessibility review
6 Communicate	Diffusion renders (SDXL + ControlNet conditioned on masterplan linework); video generation for 1-minute animation; LLM narrative editing	Final CAD linework, material palette, camera paths	Photoreal views, animation, plain-language project story	Every image audited against the plan; hallucinated elements removed before submission

B. WORKFLOW DIAGRAM



C. DESIGN ALTERNATIVES EXPLORED

From the 43 generated masterplans, the team shortlisted three families. **Central Oasis** concentrated water and shade in one large heart — dramatic, but it left the park edges thermally hostile and demanded the highest irrigation. **Braided Wadis** wove three dry-stream fingers across the site — water-wise and beautiful after rain, but its fragmented paths tested poorly for continuous comfortable walking.

Shade Loop inverted the logic: distribute comfort as an 858 m continuous shaded network — a 398 m figure-eight braided with a 460 m perimeter ring, ~1 km including the wadi spur — and hang the ten garden rooms from it. It scored highest on the two criteria the community ranked first — unbroken shade (92% of the loop shaded at 3 pm, verified 91.6%) and walkability — and the agent-based tests confirmed the loop absorbs peak Friday crowds without congestion.

CRITERION (WEIGHT)	CENTRAL OASIS	BRAIDED WADIS	SHADE LOOP
Shade continuity (30%)	6	7	9
Walkability (25%)	7	6	9
Water efficiency (20%)	4	8	7
Cost within AED 35M (15%)	6	5	7
Ten-room program fit (10%)	7	7	9
Weighted total /10	5.95	6.65	8.30

Each score is the reconciled median of three designers scoring independently, anchored to measured evidence wherever it exists: shade continuity from the 3 pm June solar study, walkability from simulated loop-completion rates, water from the irrigation model, cost from the quantity surveyor's estimate. The AI computed those metrics and flagged inconsistencies between scorers; it did not cast a vote.

D. LIMITS AND ETHICS

We state our limits plainly. The synthetic visitor personas used in agent-based simulation were audited against actual survey demographics to correct an early bias that under-represented elderly residents and domestic

workers who use the park at off-peak hours. The design deliberately contains **no biometric surveillance**: visitor counting for park management relies on anonymous, non-identifying sensors only. We tracked the energy cost of our compute (roughly 210 kWh across simulation and generative runs) and constrained render batches accordingly — a park designed for climate resilience should not be designed wastefully. Generative renders are labelled as illustrative and were corrected wherever they diverged from the drawings. Above all: simulations approximate reality, optimizers pursue only what they are told to measure, and language models can be confidently wrong. A decision log therefore records every gate in Fig. 5.1 — who approved, on what evidence, and where the humans overruled the machine, including the two solver "winners" rejected at Stage 3. Every deliverable in this entry carries a named human signature, and every design decision — from the loop's alignment to the last planting bed — was made, and can be defended, by the NABTA team.